



10 CFR 50.73

CCN: 22-89

December 15, 2022

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Peach Bottom Atomic Power Station (PBAPS) Unit 2
Subsequent Renewed Facility Operating License No. DPR-44
NRC Docket No. 50-277

Subject: Licensee Event Report (LER) 2022-002-00 Automatic Depressurization
System Safety Relief Valve Actuator Diaphragm Degraded

The subject report is being submitted in accordance with 50.73(a)(2)(i)(B) for any operation or condition which was prohibited by the plant's Technical Specifications.

There are no commitments contained in this letter. If you have any questions, please contact the Peach Bottom Regulatory Assurance Manager, Mr. Wade Scott at (717) 456-3047.

Respectfully,

A handwritten signature in black ink, appearing to read "David A. Henry", written over a horizontal line.

David A. Henry
Site Vice President
Peach Bottom Atomic Power Station

Enclosure

cc: USNRC, Administrator, Region I
USNRC, Senior Resident Inspector
W. DeHaas, Commonwealth of Pennsylvania
S. Seaman, State of Maryland
B. Watkins, PSE&G, Financial Controls and Co-Owner Affairs

**LICENSEE EVENT REPORT (LER)**

(See Page 3 for required number of digits/characters for each block)
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Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk air: omb_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Peach Bottom Atomic Power Station, Unit 2 (PB2)

2. Docket Number

05000 277

3. Page

1 OF 4

4. Title

Automatic Depressurization System Safety Relief Valve Actuator Diaphragm Degraded

5. Event Date

Month	Day	Year
10	17	2022

6. LER Number

Year	Sequential Number	Revision No.
2022	- 002 -	00

7. Report Date

Month	Day	Year
12	15	2022

8. Other Facilities Involved

Facility Name	Docket Number
	0500
Facility Name	Docket Number
	0500

9. Operating Mode

4

10. Power Level

00

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)**10 CFR Part 20**☐ 20.2203(a)(2)(vi)☐ 50.36(c)(2)☐ 50.73(a)(2)(iv)(A)☐ 50.73(a)(2)(x)☐ 20.2201(b)☐ 20.2203(a)(3)(i)☐ 50.46(a)(3)(ii)☐ 50.73(a)(2)(v)(A)**10 CFR Part 73**☐ 20.2201(d)☐ 20.2203(a)(3)(ii)☐ 50.69(g)☐ 50.73(a)(2)(v)(B)☐ 73.71(a)(4)☐ 20.2203(a)(1)☐ 20.2203(a)(4)☐ 50.73(a)(2)(i)(A)☐ 50.73(a)(2)(v)(C)☐ 73.71(a)(5)☐ 20.2203(a)(2)(i)**10 CFR Part 21**☒ 50.73(a)(2)(i)(B)☐ 50.73(a)(2)(v)(D)☐ 73.77(a)(1)(i)☐ 20.2203(a)(2)(ii)☐ 21.2(c)☐ 50.73(a)(2)(i)(C)☐ 50.73(a)(2)(vii)☐ 73.77(a)(2)(i)☐ 20.2203(a)(2)(iii)**10 CFR Part 50**☐ 50.73(a)(2)(ii)(A)☐ 50.73(a)(2)(viii)(A)☐ 73.77(a)(2)(ii)☐ 20.2203(a)(2)(iv)☐ 50.36(c)(1)(i)(A)☐ 50.73(a)(2)(ii)(B)☐ 50.73(a)(2)(viii)(B)☐ 20.2203(a)(2)(v)☐ 50.36(c)(1)(ii)(A)☐ 50.73(a)(2)(iii)☐ 50.73(a)(2)(ix)(A)☐ OTHER (Specify here, in abstract, or NRC 366A).**12. Licensee Contact for this LER****Licensee Contact**

Wade J. Scott, Regulatory Assurance Manager

Phone Number (Include area code)

7174563047

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
X	SB	RV	T020	Y					

14. Supplemental Report Expected☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)**15. Expected Submission Date**

Month

Day

Year

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On October 17, 2022, with the unit in a refueling outage. Water was identified leaking from a fastener on the PB2 RV-71B Main Steam Safety/Relief Valve (S/RV) body. The water leak was a small steady stream. The leakage was due to a degraded gasket and steam cut fastener threads. Subsequent laboratory analysis of the leaking valve identified the Automatic Depressurization System, (ADS) function performed by the pneumatic operator was non-functional. The pneumatic operator resilient components (diaphragm) were discovered to be embrittled and lost pressure retaining integrity. The timeframe for the degraded ADS capability cannot be reasonably determined at this time. The relief valve overpressure function was evaluated to be not impacted by the leak or the loss of ADS function. Therefore, Peach Bottom is submitting this report in accordance with 10 CFR 50.73(a)(2)(i)(B), as any operation or condition which was prohibited by the plant's Technical Specifications, having exceeded the required 14 day action time or maximum Risk Informed Completion Time (RICT) to restore an ADS valve to OPERABLE status. There were no other systems which were inoperable during this identified condition that contributed to this event. There were no actual safety consequences as a result of this event.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Peach Bottom Atomic Power Station, Unit 2	05000- 277	2022	- 002	- 00

NARRATIVE**Plant Operating Conditions before the Event:**

Unit 2 was in Mode 4, Cold Shutdown, with the Main Steam Lines water solid, operating at 0% of rated thermal power on October 17, 2022. Unit 2 was shut down for its 24th Refueling Outage (P2R24) on October 16, 2022.

Status of structures, components, or systems that were inoperable at the start of the event and that contributed to the event:

There were no structures, systems, or components (SSCs) whose inoperability contributed to this event.

The Energy Industry Identification System component function identifier and name of each component or system:

EIIS: Safety Relief Valves [SB],
RV-2-02-071B, B SAFETY RELIEF VALVE (RV-71B)
Target Rock Model 67F pilot operated safety/relief valve
Serial Number 024

Description of Event:

On October 17, 2022, at 1337 hours, station personnel identified a small steady stream of water, approximately 0.2 gallons per minute leaking from the insulation around the '71B' Main Steam/Relief Valve (S/RV). Subsequent investigation revealed that the leakage was from the relief valve pilot filter plug threaded connection. There was no through-wall leakage of any Reactor Coolant System (RCS) pressure boundary component. The RV-71B was emergently replaced in the scope of the planned outage, quarantined, and shipped to an off-site vendor for analysis and repair.

As part of vendor testing the relief valve, the pneumatic operator is functionally tested with air. The pneumatic operator failed to actuate and air leakage was audible during testing, indicating potential resilient parts degradation. Disassembly and examination of the pneumatic operator revealed the actuator diaphragm elastomer had embrittled and delaminated, enabling gross leakage that inhibited the S/RV manual operation capability. This as-found condition represents a loss of the valve's Automatic Depressurization System (ADS) function.

Technical Specification (TS) 3.5.1 Limiting Condition of Operation (LCO) requires each Emergency Core Cooling System (ECCS) injection/spray subsystem and the Automatic Depressurization System (ADS) function of five (5) S/RVs to be OPERABLE in operating modes 1, 2 and 3.

The remaining four (4) S/RVs that support ADS capability were verified to be operable for this function. Peach Bottom TS Limiting Condition for Operation 3.5.1, Condition E for one (1) ADS valve inoperable, has a Completion Time of 14 days or in accordance with the Risk Informed Completion Time Program. The event start time for the loss of ADS function of the RV-71B is indeterminate. The S/RV ADS function is considered inoperable for a period greater than permitted by TS since the start of the degradation cannot be reasonably determined. Therefore, the condition is reportable under criterion 10CFR 50.73(a)(2)(i)(B) as any operation or condition which was prohibited by the plant's Technical Specifications.

The RV-71B was installed as part of the planned preventive maintenance relief valve outage work scope in October of 2018 (P2R22). The valve was in-service approximately four (4) years at the time of discovery.

The RV-71B ADS Safety Relief Valve was not actuated during Cycle 24 operations (October 2018 – October 2020). The RV-71B was used for manual pressure control during Cycle 25 (October 2020 – October 2022) as part of post U2 SCRAM actions on May 15, 2022 (ML22196A020). Control room operators manually operated the valve from the main control room and observed the expected response.

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Peach Bottom Atomic Power Station, Unit 2	05000- 277	2022	- 002	- 00

NARRATIVE**Analysis of Event:**

The ADS system is an Emergency Core Cooling System (ECCS) which is designed to provide depressurization of the reactor coolant system during a small break loss of coolant accident if high pressure injection fails or is unable to maintain required water level in the reactor vessel. It reduces the reactor pressure vessel pressure to within the operating pressure range of the low pressure ECCS subsystems so that these subsystems can provide coolant inventory makeup. To accomplish this function, pneumatic power is used to actuate the designated SRVs A, B, C, G and K.

This failure only impacts the RV-71B ADS valve. The effect of this condition is that one (1) of five (5) ADS valves was not considered operable as required by Technical Specifications (TS) Section 3.5.1. The reduction in capability of the ADS safety function, which is needed to mitigate the consequences of an accident, has been evaluated and a single inoperable ADS S/RV does not result in loss of the safety function. The operability of the remaining four (4) ADS valves provides relative assurance that required safety functions would be accomplished as designed and therefore this event is not reportable under 50.73(a)(2)(v)(D). This event is not an unanalyzed condition and therefore not reportable under criterion 10CFR 50.73(a)(2)(ii)(B).

The RV-71B over-pressure protection feature has been evaluated and was not adversely impacted by the leakage from the pilot filter plug threaded connection. The volume of the leak through the pilot filter fastener threads in comparison to pilot sensing drilled passage volume is minute. The relief valve pilot assembly would have been subjected to uninterrupted or undiminished main steam process pressure, thus the valve operation set point was not impacted. Technical Specification 3.4.3, Safety Relief Valves (SRVs) and Safety Valves (SVs), Limiting Condition of Operation requires the safety function of 13 valves (any combination of SRVs and SVs) to be OPERABLE in modes 1, 2, and 3. RV-71B was installed as 1155 +/-11.5 PSIG setpoint valve, the highest of three (3) groups of S/RVs. This function was not adversely impacted by this event. There was no loss of over-pressure protection safety function.

No other systems or secondary functions were affected. There were no automatic or manual safety system responses associated with this event. There were no operator actions associated with this event. Based on the above, sufficient systems were available to provide the required safety functions needed to protect the health and safety of the public.

Cause of the Event:

The RV-71B pneumatic operator failure proximate cause is thermal degradation of its resilient components. The prolonged exposure to elevated temperatures was the result of a steam leak in its immediate vicinity.

Main Steam S/RVs are insulated by thermal blankets encased in a removable stainless steel metal enclosure. The relief valve main body inlet, outlet and pilot assembly are all enclosed in the insulation blankets and enclosure. These parts generally operate at or very near main steam process temperature. The relief valve pneumatic operator is not insulated and protrudes through the sheet metal enclosure into the drywell atmosphere. This configuration reduces the steady state temperature exposure of the pneumatic operator resilient components.

The identified leakage from the pilot filter fastener has been evaluated by station Engineering staff and concluded to have no adverse impact on the RV-71B over-pressure protection function. The pilot filter plug fastener is removed inspected and reinstalled during every relief valve refurbishment activity. Widening of the thread gaps from wear enabled steam cutting that compromised the pilot filter fastener outer gasket. The fastener threads are believed to have slightly worn over the operating life of this valve.

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NARRATIVE

Leakage past the fastener threads compromised the pilot filter plug outer gasket enabling a flow path for main steam to enter into the drywell. The small volume steam leak impinged on the relief valve insulation blankets and wet vapor flow exited the metal enclosure opening through which the pneumatic operator protrudes. The wet vapor flow exiting the metal enclosure aperture subjected the pneumatic operator to elevated operating temperatures. Over some indeterminate time period following its last observed operation in May 2022, the operator diaphragm was adversely impacted. The diaphragm as-found condition was typical of sustained exposure to temperatures above its material qualifications. No human performance related causes have been identified.

Corrective Actions:

RV-71B was replaced in kind with a refurbished, tested and fully qualified replacement S/RV prior to unit start up from the refueling outage (P2R24) and entry into the mode of applicability for TS 3.4.3 and 3.5.1.

All other S/RVs were visually inspected for proper insulation installation and any evidence of steam leaks from their respective pilot filter fasteners and other bolting. In addition, all S/RV pneumatic operators were tested in-situ to validate full function and integrity of the pneumatic operator resilient parts. S/RVs replaced within the planned maintenance scope of the refueling outage (RV-71G, , RV-71J, RV-71K) also underwent review of the refurbishment and testing work package documentation to ensure pilot filter fastener and gasket tolerances were within design limits. No further deficiencies were identified.

The failed resilient parts shall undergo analysis to ensure compliance with applicable design specifications. Causal evaluation results will be reviewed by refurbishment vendor and the original equipment manufacturer.

Previous Similar Events:

11-003-00 Automatic Depressurization System Safety Relief Valve Actuator Diaphragm Thread Seal Leak (ML11325A383)